

Climbing to the Frontier — RESULTS

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In one line. We hand in a write-up for all 24 cells. 16 cells are complete proofs (P2 C5 counts here: both of its bounds are proved), 6 are partial and say exactly which part is proved, and 2 (P4 C4/C5) are computational certificates. Every cell file states *solved* or *partial* on its first page. Anything proved only with a computer is labelled *computer-assisted* and ships with its code and log.

Cell	Status	What we claim
P1 — Angles between lines (Fejes Tóth)		
C1	solved	Planar case, self-contained proof (the result itself is known [FVZ16, BM19]).
C2	solved	The orthogonality lemma.
C3	solved	$d + 1$ lines in $\mathbb{R}^d$, via a saturation step and a matching argument.
C4	solved	5 lines in $\mathbb{R}^3$, 6 in $\mathbb{R}^4$. The 6-line case is also given a second proof via the C5 cycle lemma.
C5	solved	$N = d + 2$ lines for <i>every</i> d . New ingredient: a cycle lemma (tridiagonal/cyclic nullity plus a monotone potential), checked numerically on $4 \cdot 10^4$ cycles.
C6	partial	$N = 6$ lines in $\mathbb{R}^3$ (new instance, by subset averaging). We prove that averaging from all known cases gives <i>no</i> infinite family beyond it. Structural reduction of $N = d + 3$ for all d : everything is settled except components of rank ≥ 4 , nullity ≥ 3 and cyclomatic number ≥ 2 (a new PSD forcing lemma). First open case: 7 lines in $\mathbb{R}^4$.
P2 — Uphill paths on the hypercube		
C1–C4	solved	$U(Q_3) = 14$, $U(Q_4) = 34$, $U(Q_5) = 88$, $U(Q_6) = 204$, $U(Q_7) = 464$, $U(Q_8) = 1040$, all <i>with proofs of optimality</i> . Key new tool: an exact formula $P(f) = V + \sum_{u \in R} (\deg u - 1) + \sum_{u \in R} (N(u) - 2) \text{out}(u)$, which gives $U(G) \geq V + (d-1)\nabla(G)$ for d -regular G (∇ = decycling number). Code labellings give the matching upper bound. $\nabla(Q_5) \geq 14$ and $\nabla(Q_6) \geq 28$ are SAT facts, each checked with two solvers and by a second, independently written program.
C5	solved (bounds)	2328 $\leq U(Q_9) \leq 2400$. The lower bound is new: it beats the counting bound 2312, using a split of Q_9 into eight Q_6 copies, a parity lemma (SAT) and a Delsarte LP bound. The upper bound comes from a 20-word even code.
C6	partial	Reduction: beating 2400 requires a decycling set of Q_9 of size ≤ 235 . Conjecture: $U(Q_9) = 2400$. A direct SAT search for such a set ran for about 3 hours without an answer.
P3 — Bulgarian solitaire		
C1, C2	solved	Cyclic partitions and cycles; $D_B(T_k)$ (reproduces Griggs–Ho, with complete proofs).
C3	partial	The general bound and $D_B(T_k - 1)$ are proved. The extremal set is given only by an exact dynamical criterion.
C4	solved	$D_B(T_{k-1} + 1) = (k-1)(k-3)$ for all $k \geq 5$, with no computation. This is the $r = 1$ case of Griggs–Ho Conjecture 4.7, which we did not find proved in the literature we checked.
C5	partial	$D_B(T_{k-1} + 2) = (k-1)(k-4)$ for all $k \geq 7$ by one uniform argument (Conj. 4.7, $r = 2$), and values 2, 3, 5, 8, 12 for $k = 2..6$. One explicit extremal family is given; the full extremal set only via a criterion.
C6	partial	New: $d_B(1^n) = n - k + 1$ for every n , with the exact landing cycle, which proves the lower bound of Conj. 4.7 in the two middle columns. $D_B(n)$ computed exactly for all $n \leq 80$ (300 s): it equals Conj. 4.7 everywhere, with no disagreement. Four columns ($r = 1, 2, k-1, k$) are proved in C2–C5.
P4 — Disjoint congruence classes (Sun’s conjecture)		
C1–C3	solved	$k = 3$, $k = 4$, $k \leq 8$ (known from O’Bryant; independent proofs).
C4	solved	All $k \leq 12$. For each $9 \leq k \leq 16$ we report exactly what is left undecided, which is nothing: every survivor is decided UNSAT.
C5	solved (cert.)	Certified boundary $K^* = 14$, meeting requirements (i)–(v): the vacuity test finds genuine systems; every survivor has an individual decision and its minimal forcing part (files attached); exact survivor lists; the extra pruning rule is proved and switching it off leaves the lists unchanged; two independent implementations (SAT and DFS) give elementwise-identical lists. Timed reruns take 193s at $k = 14$ but 613s at $k = 15$, just over the 10-minute rule, so $k = 15, 16$ are reported as <i>decided beyond the certified boundary</i> (all UNSAT, no disagreement). $k = 24$ and $k = 30$ are decided by hand (the prime $k-1$ argument).
C6	partial	Part (a), “decide a size $k \geq 25$ ”: $k = 32, 38$ and 42 are decided, computer-assisted (O’Bryant’s method with new ingredients, finishing with an exhaustive 4s check over a finite set of prime distributions, rerun reproduced). $k = 44$ is computer-assisted but the certificate is incomplete (runtime estimate ≈ 240 CPU-hours). $k = 48$ is open.

Cells 5 and 6. All four C5 cells carry a real result: P1 C5 for all d ; a new lower bound in P2 C5; the $r = 2$ case

of the Griggs–Ho conjecture in P3 C5; and a certified boundary with two independent implementations in P4 C5. We hand in partial progress on all four open C6 cells.

What we do *not* claim. We do not claim the full Fejes Tóth conjecture, the exact value of $U(Q_9)$, the full function $D_B(n)$, $k \geq 15$ in the certified range, or $k \geq 44$ for Sun’s conjecture. “New” means new relative to the sources we cite (Griggs–Ho 1998, O’Bryant 2007, Bilyk–Matzke 2019); we did not survey the literature exhaustively.

Where to find it. One folder per cell (`Pi-Cj/`) with the PDF, the L^AT_EX source, code and logs; `INDEX.md` lists the file handed in for every cell.